A micro Lie theory for state estimation in robotics

Solà, Deray and Atchuthan, arXiv:1812.01537v9, pp. 1–17

Lesson t2.6 · Exp, Log, $\oplus$, $\ominus$, Ad and the Jacobians a robot needs

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Why robots need this

- A rotation, a pose and a unit quaternion all live on a **curved** space, not in $\mathbb{R}^n$
- Adding an increment to a quaternion leaves the unit sphere, so it gets renormalized by hand
- A covariance on a 9-number rotation matrix is singular: the state has 3 degrees of freedom
- Every Jacobian in an EKF or a graph solver needs a definition that respects the curvature
- **One idea**: do all calculus in the flat tangent space $\mathbb{R}^m$, and move only through Exp and Log

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Roadmap

1. The group, Exp and Log
2. Plus, minus and the adjoint
3. Jacobians and the chain rule
4. Covariance and integration
5. The ESKF on SE(2)
6. Traps and what comes next

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The one idea

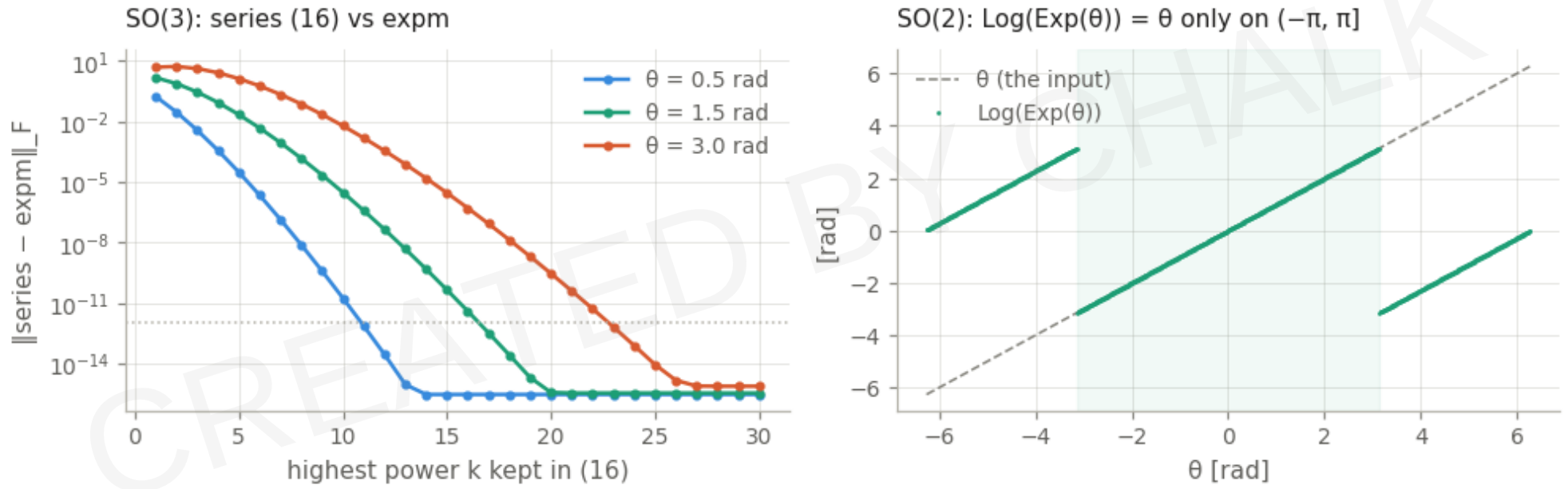
Work in the tangent space $\mathbb{R}^m$; Exp and Log are the only way on and off the group

Plus and minus then make a curved state read like a vector: state $\oplus$ increment, state $\ominus$ state

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Exp wraps the tangent line onto the group

Fig L4 · (14)–(24), Ex. 4 Exp is a convergent series; Log returns the principal angle



Left: the Exp series for SO(3) against expm. Right: Log wraps at ±π

Log(Exp(θ)) returns θ only on (−π, π]; outside it wraps

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Plus and minus: the increment lives in the local frame

$$\mathcal{Y} = \mathcal{X} \oplus \boldsymbol{\tau} \triangleq \mathcal{X} \circ \mathrm{Exp}(\boldsymbol{\tau}) \tag{25}$$

$$\boldsymbol{\tau} = \mathcal{Y} \ominus \mathcal{X} \triangleq \mathrm{Log}(\mathcal{X}^{-1} \circ \mathcal{Y}) \tag{26}$$

$\mathcal{X}, \mathcal{Y}$ elements of the group

$\boldsymbol{\tau}$ tangent vector, m numbers

Exp, Log $\mathbb{R}^m \rightleftarrows$ group (21)–(24)

Right $\oplus$ and $\ominus$ are used throughout; the left ones (27)–(28) are in the global frame

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Predict: right or left $\oplus$?

- Same pose $\mathcal{X}$ and same $\tau = (1.2\text{ m}, 0\text{ m}, 0.9\text{ rad})$
- One robot applies τ with right-$\oplus$, another with left-$\oplus$

? **Do the two end positions coincide?**

A Yes, $\oplus$ is commutative

B Only if the heading is zero

C No, they differ by tens of centimetres

D No, they differ by a full turn

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The adjoint converts local increments to global ones

$$ {}^{\mathcal{E}}\boldsymbol{\tau} = \mathbf{Ad}_{\mathcal{X}}\, {}^{\mathcal{X}}\boldsymbol{\tau} \tag{30} $$

$$ \mathcal{X} \oplus \boldsymbol{\tau} = (\mathbf{Ad}_{\mathcal{X}}\boldsymbol{\tau}) \oplus \mathcal{X} \tag{32} $$

Ad_$\mathcal{X}$ m×m matrix

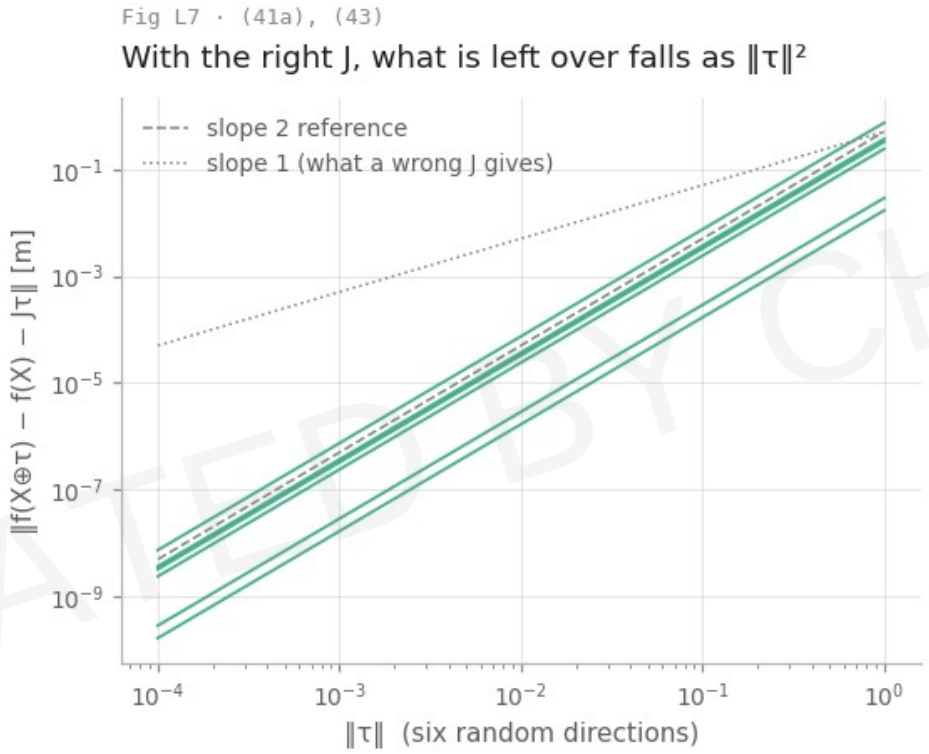
${}^{\mathcal{X}}\boldsymbol{\tau}$ increment in the frame at $\mathcal{X}$

${}^{\mathcal{E}}\boldsymbol{\tau}$ increment in the global frame

For rotations Ad_R = R; for poses it also mixes in the lever arm

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A Jacobian on a group passes the slope test



$\|f(\mathcal{X} \oplus \tau) - f(\mathcal{X}) - J\tau\|$ against $\|\tau\|$ in six random directions

A correct J has slope 2 here; a wrong one leaves slope 1

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Every Jacobian is a product of four blocks

1 Inverse

J of X^{-1} with respect to X is $-Ad_X$ (62)

2 Composition

J of $X \circ Y$ is $Ad_{Y^{-1}}$ in X and I in Y (65), (66)

3 Exp

$J_r(\tau)$ in (67); $Ad_{Exp(\tau)} = J_l J_r^{-1}$ (75)

4 Action

Per group, App. A–E; for SO(3) it is $-R[v]_\times$ (150)

5 Chain rule

J of Z with respect to X is the product of the factors (58)

Test the whole Jacobian once, then each block on its own

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Covariance on the tangent is first order only

73.9 %

of the cloud inside the 2σ ellipse

86.5 %

expected for a true Gaussian

2.91 m

cloud mean, against 3.00 m

- Σ is m×m on the tangent: 3×3 for SE(2), 6×6 for SE(3)
- Propagation is $J \Sigma J^T$ (55); the banana appears once angle and lever arm grow together

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Integrate with Exp, not Euler

4.4×10^{-16} m

Exp integration, δt = 1.5 s, 9 s

1.18 m

Euler on (x, y, θ), same run

0.078 m

Euler at δt = 0.1 s

- X_k = X_{k−1} Exp(v_k δt_k) (56), exact for a piecewise-constant twist

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The ESKF on SE(2): two lines change

1 Predict mean

$\hat{X} \leftarrow \hat{X} \oplus u$ (99)

2 Predict P

$P \leftarrow F\,P\,F^T + G\,W\,G^T$, $F = \mathrm{Ad_Exp}(u)^{-1}$, $G = J_r(u)$ (100)

3 Innovate

$z = y - \hat{X}^{-1} \cdot b$; $Z = H\,P\,H^T + N$; $K = P\,H^T\,Z^{-1}$

4 Update mean

$\hat{X} \leftarrow \hat{X} \oplus Kz$ (101)

5 Update P

$P \leftarrow P - K\,Z\,K^T$ (102)

Compared with the textbook EKF, only + became $\oplus$ and the Jacobians come from the blocks

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Predict: a robot standing still

- Printed J_r of SO(3) in (143)
- Gyro samples at the IMU rate, $\theta \approx 0$ on every one

? **What does the coefficient $(1 - \cos\theta)/\theta^2$ return in float64 for tiny θ?**

A The exact value 0.5

B A slightly noisy 0.5

C 0, losing every digit

D Infinity

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Symptom to fix

Symptom	Cause	Fix
FD slope 1, not 2	Wrong block or wrong $\ominus$ side	Test each block alone
NaN when standing still	Divide by θ in Exp, Log, J_r	Taylor branch near 0
Turns the long way	Log branch cut at π	Flip q to $w > 0$; wrap
Log(R) wrong near 180°	(135) is 0/0 at π	Use symmetric part
Overconfident when turning	Euler step, or F from (x, y, θ)	Exp step; $F = \mathrm{Ad}^{-1}$
Wrong in world frame	Frame mismatch	${}^{E}\Sigma = \mathrm{Ad}\ \Sigma\ \mathrm{Ad}^{T}$ (54)
Codebases disagree	Tangent order or side	Permute blocks; (75)

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What to be able to defend

- State $\oplus$ increment and state $\ominus$ state give exactly dof numbers, so covariances have the right size
- Right $\oplus$ puts the increment in the body frame; Ad_X moves it to the world frame
- Every Jacobian is a chain of inverse, composition, Exp and action blocks, checked by finite differences
- Integrate with X Exp(v δt); Euler on (x, y, θ) drifts in every turn
- Every closed form divides by θ, θ^2 or θ^3: write the small-angle branch first

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What comes next

- EKF localization (Thrun ch. 7): §V-A with the Jacobians written once in Lie form
- Graph SLAM (Grisetti, t2.8): $\oplus \diamondsuit$ and $\ominus \diamondsuit$ on a composite state; FAST-LIO2 is such a composite
- IMU preintegration (Forster et al., t2.11): built on (56), (68)–(70) and J_r
- Barfoot ch. 8: the left-perturbation view; (75) and (147) are the bridge

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